



Module 5: The Meat of Rust

Lesson 5.1: Structs

Structs are used in Rust to define custom data types with fields, and unlike classes in other languages, they do not support inheritance. Struct behavior (methods and associated functions) is defined in a separate `impl` (implementation) block.

Key Takeaways

- Instantiation requires specifying a value for every field, but shorthand initialization can be used if variable and field names match.
- Associated functions are defined within the `impl` block without a `self` parameter; they often act as constructors and are conventionally named `new`.
- Methods always take some form of `self` (e.g., `&self` or `&mut self`) as their first parameter.
- The scope operator (`::`) is used to access associated functions (e.g., `RedFox::new()`).

Key Concepts

- **Struct:** A custom data structure used to group related data fields, utilizing Capital Camel Case naming convention.
- **Implementation Block (`impl`):** The required section where a struct's associated functions and methods are defined.
- **Associated Function:** A function tied to the struct type itself (not an instance), often used for creation (`new`).
- **Method:** A function called on an instance of the struct, always taking a form of `self` as the first argument.



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Lesson 5.2: Exercise H - Structs

Exercise H guided the creation of a Polygon struct, including implementing an associated function (new) that provided default field values. It demonstrated creating methods that take `&self` for reading and `&mut self` for modification (increment_sides) and utilizing shorthand initialization when instantiating the struct.

Key Takeaways

- Self (capital S) within an impl block is idiomatic syntactic sugar for the struct's name.
- Methods that need to mutate the struct's internal data must accept a mutable reference to self (`&mut self`).
- Struct initialization is made cleaner by using shorthand when variable names match field names.
- Match expressions are effective inside methods (like `shape`) to return different values based on the struct's internal state (`self.sides`).

Key Concepts

- **Struct Shorthand:** A method of instantiation where `field_name: variable_name` is shortened to just `field_name` if the names are the same.
- **Mutable Method:** A method requiring `&mut self` as its first parameter to gain exclusive access necessary for modifying the struct's fields.



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Lesson 5.3: Traits

Traits define required behavior, similar to interfaces, and represent Rust's preference for composition over inheritance. Once a type implements a trait, generic functions can accept any value that satisfies that trait, enabling flexible and powerful generic programming.

Key Takeaways

- Traits can be implemented for any type—even built-in primitives (like `u8`) or types from external packages—provided either the trait or the type is defined in the current package.
- The special `Copy` trait dictates that a value should be copied instead of moved; it is reserved for small values that reside entirely on the stack.
- Traits can inherit from other traits, creating a hierarchy where implementing a sub-trait mandates implementing its super-traits as well.
- Traits can provide default method behaviors that structs can choose to override or use as is.

Key Concepts

- **Trait:** A mechanism for shared behavior; defining methods that a type must implement to gain specific functionality.
- **Composition over Inheritance:** Rust's architectural approach using traits to share behavior, avoiding the complexity of class inheritance hierarchies.
- **Generic Functions:** Functions that use generics to accept any type `T` constrained by the requirement that `T` implements a specific trait.
- **Default Behavior (Trait):** Providing a standard implementation for a trait method directly within the trait definition, which can be overridden by the implementing type.



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Lesson 5.4: Exercise I - Traits

Exercise I involved defining and implementing the Colorful trait for a custom struct (Hat) and the primitive type i32. It also demonstrated refactoring a standalone helper function (isEven) into an idiomatic trait method (is_even on the EvenOdd trait), simplifying the calling syntax thanks to the dot operator's auto-dereferencing feature.

Key Takeaways

- A match expression is a good tool inside a trait method implementation to return different values based on internal state.
- The `impl Trait for Type` syntax works for built-in types like `i32` as well as custom structs.
- Refactoring a function into a method eliminates the need for manual dereferencing of `self` when accessing the value.

Key Concepts

- **impl Trait for Type:** The syntax used to bind a specific set of behaviors defined by a trait to a data type.
- **Refactoring to Idiomatic Method:** Converting external helper functions into methods defined by a trait to utilize Rust's self reference passing and auto-dereferencing.



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Lesson 5.5: Collections

Collections are generic types in the standard library, with `Vector (Vec<T>)` being the most common, functioning as a dynamically sized list of a single type. `HashMap (HashMap<K, V>)` is another key collection, providing constant-time insertion, lookup, and removal of key-value pairs.

Key Takeaways

- Vectors store objects contiguously, allowing efficient indexing, but Rust panics (crashes gracefully) if an index is out of bounds.
- The `vec!` macro is the preferred, ergonomic way to initialize a vector with literal values.
- `HashMap` retrieval methods often return the `Option` enum, indicating whether the key was present (`Some`) or absent (`None`).
- Specialized collections include `VecDeque` (double-ended queue) and `BinaryHeap` (priority queue that pops the max value).

Key Concepts

- **Vector (`Vec<T>`):** A generic, dynamically sized collection holding elements of a single type in continuous memory.
- **HashMap (`HashMap<K, V>`):** A key-value data structure optimized for quick (constant-time) access and modification.
- **`vec!` Macro:** The macro used for concise initialization of vectors with pre-defined literal values.
- **Option Return:** The standard way collections signal the possibility of a value being absent during retrieval.



Lesson 5.6: Exercise J - Collections

Exercise J provided practical application of vectors, demonstrating how to initialize them both through the verbose `Vec::new().push()` method and the more concise and performant `vec!` macro. The exercise reinforced that vectors own their values, requiring the vector variable to be mutable for operations like sorting (`.sort()`) or pushing new elements.

Key Takeaways

- The `vec!` macro is generally preferred for vector initialization due to its conciseness and compiler optimization opportunities.
- When an item is pushed into a vector, the value is moved, and the original variable loses ownership and is invalidated.
- Methods that modify the vector's contents or order (like `.sort()` or `.push()`) require the vector to be declared with `let mut`.
- A `for` loop provides a straightforward way to iterate over and potentially consume the items in a vector.

Key Concepts

- **vec! Macro Initialization:** The most ergonomic way to create a vector and populate it with literals in a single line.
- **Vector Mutability:** Required for in-place modifications such as adding elements (`push`) or sorting the contents (`sort`).
- **Consumptive Iteration:** Iterating over a vector using `for item in vector` takes ownership of the items, preventing further use of the vector.



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Lesson 5.7: Enums

Rust enums are powerful algebraic data types that can associate data (tuples or structs) with their variants, providing functionality beyond simple C-like enums. The standard method for inspecting and acting upon enum variants is the `match` expression, which requires exhaustive handling of every possible variant.

Key Takeaways

- The generic enum `Option<T>` (`Some(T)` or `None`) is standard across the library and is used whenever a value might be absent.
- The generic enum `Result<T, E>` (`Ok(T)` or `Err(E)`) is Rust's primary mechanism for error handling, replacing exceptions.
- The compiler enforces that a `Result` must be used (not dropped silently), encouraging programmers to address all possible errors.
- `if let` expressions are useful when checking for a single variant match, while `match` expressions are required for handling all possibilities.

Key Concepts

- **Enum (Algebraic Data Type):** A custom type whose variants can carry associated data of different types, behaving like a tagged union.
- **Match Expression:** A control flow structure similar to a switch statement that demands all possible enum variants are handled (exhaustiveness).
- **Option<T>:** The standard library enum representing presence or absence of a value, often used as a replacement for null.
- **Result<T, E>:** The enum used for I/O and operations that might fail, requiring conscious error handling (e.g., using `.unwrap()` or pattern matching).